

Design & Analysis of Algorithms - Capstone Assignment (Solved)

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Course: Design and Analysis of Algorithms Lab Faculty: Dr. Aarti

Source assignment (for problem statement & sample data): uploaded 'Capstone Assignment.pdf'.

Reference in chat: [filecite turn0file0](#)

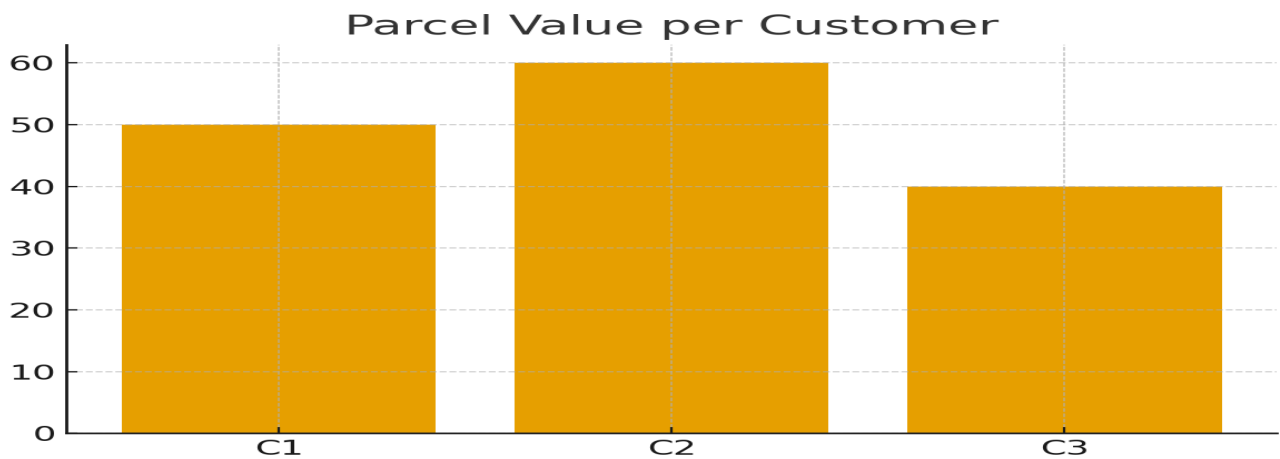
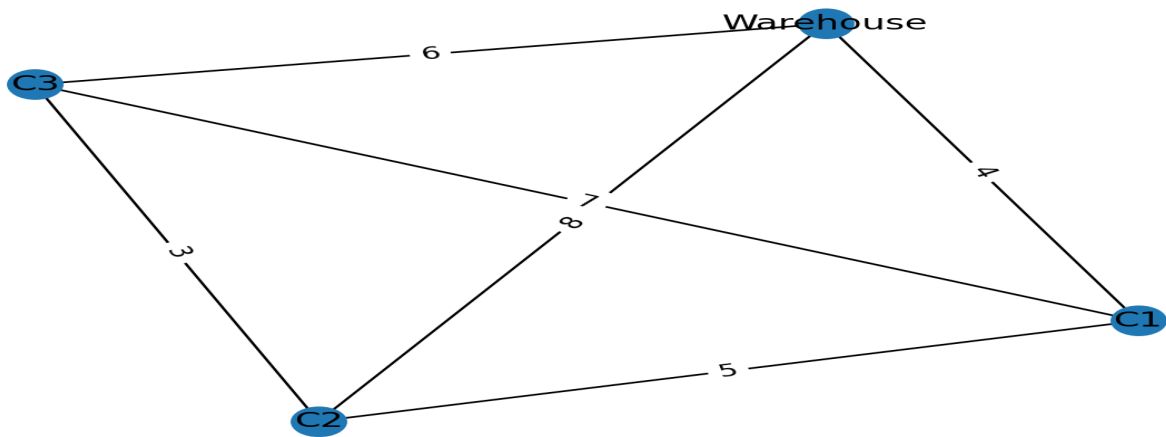
Problem Summary:

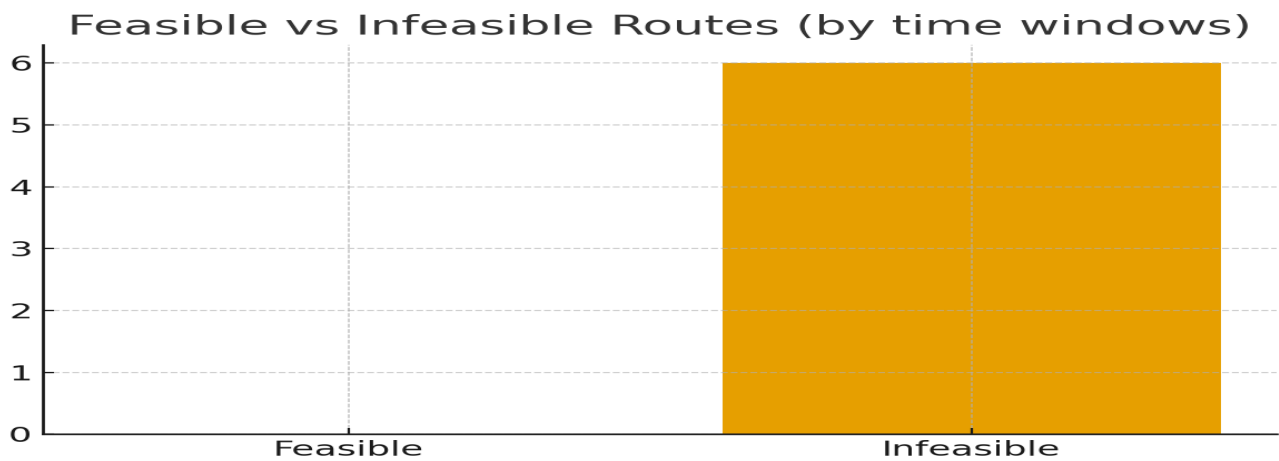
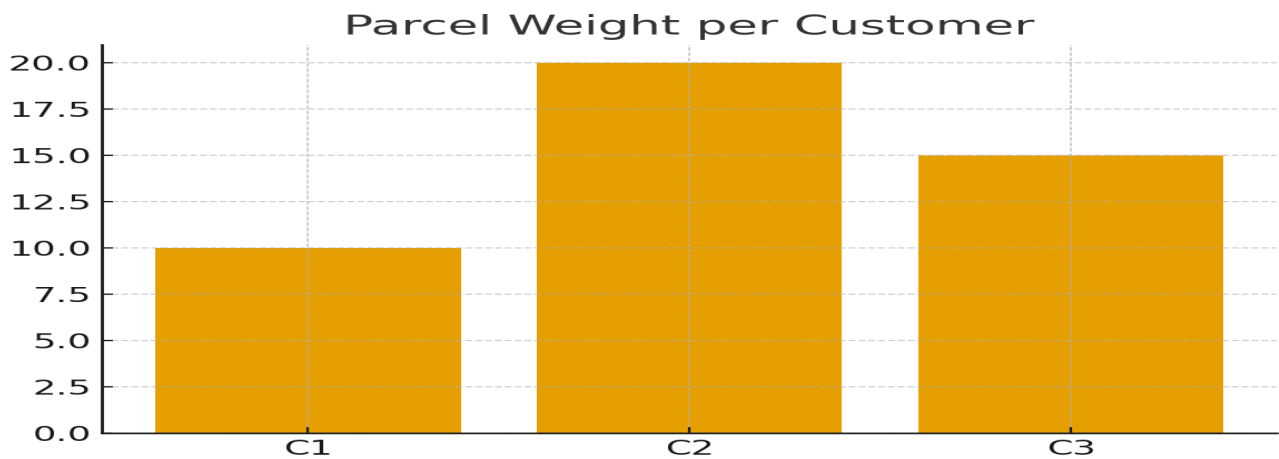
Optimize a delivery route from a warehouse to customer locations to minimize travel distance/time, respect delivery time windows, maximize parcel value under vehicle capacity.
Algorithms used: Recurrence, Greedy (0/1 knapsack heuristic), Dynamic Programming (time-window check), Graph algorithms (Dijkstra, Prim), and TSP brute-force for small instances.

Key Results:

- Recurrence-based example cost (recursive TSP estimation starting at Warehouse): 18
- Greedy parcel selection (by value/weight): chosen=['C1', 'C2'], total_value=110, total_weight=30
- Feasible routes respecting time windows (start at 9): 0 found
- Dijkstra distances from Warehouse: [0, 4, 8, 6]
- Prim MST edges: [('Warehouse', 'C1', 4), ('C1', 'C2', 5), ('C2', 'C3', 3)]
- TSP (brute force) best route: ['Warehouse', 'C1', 'C2', 'C3', 'Warehouse'], cost: 18

Plots and Visuals:





Algorithm Snippets and Explanations:

- 1) Recurrence: A recursive TSP-like function explores next unvisited city and returns min total cost.
- 2) Greedy: Select parcels by highest value/weight ratio until capacity exhausted (heuristic).
- 3) DP for time-windows: We check permutations for feasibility under delivery time windows (exact for small n).
- 4) Dijkstra: Single-source shortest paths from warehouse to each customer (used for travel time estimates).
- 5) Prim MST: Connect all nodes minimizing total connecting distance (useful when no return required).
- 6) TSP Brute-force: Permute all customer visit orders, compute round-trip cost, select minimum (exponential).

Notes: For full submission to LMS/GitHub, include a Jupyter notebook (delivery_route_optimization.ipynb), README, and this report.

This solution was prepared automatically using the sample data from the uploaded assignment (Capstone Assignment.pdf).